Pharmacokinetics/ Pharmacodynamics

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Learning Objectives

1. Describe the changes in critically ill patients that alter drug absorption.
2. Explain how critical illness affects drug distribution.
3. Depict the effects of changing hepatic blood flow, intrinsic activity, and protein binding on drug metabolism.
4. Differentiate between different critically ill patient populations and the expected pharmacokinetic (PK) changes.
5. Identify the desired pharmacodynamic variables associated with efficacy in select drugs.

Abbreviations in This Chapter

AAG	α_1-Acid glycoprotein
AKI	Acute kidney injury
ARC	Augmented renal clearance
AUC	Area under the curve
AUC/MIC	Ratio of area under the curve to the minimum inhibitory concentration for the bacterial pathogen
$AUC_{0\text{-}24}$/MIC	Ratio of area under the curve for 24 hours to the minimum inhibitory concentration for the bacterial pathogen
CKD	Chronic kidney disease
$C_{max/MIC}$	Ratio between the maximum drug concentration and the minimum inhibitory concentration for the bacterial pathogen
fT>MIC	Free drug concentration time above the minimum inhibitory concentration for the bacterial pathogen
ECMO	Extracorporeal membrane oxygenation
GFR	Glomerular filtration rate
ICU	Intensive care unit
MIC	Minimum inhibitory concentration
PD	Pharmacodynamic(s)
PK	Pharmacokinetic(s)
TBI	Traumatic brain injury
TDM	Therapeutic drug monitoring
Vd	Volume of distribution

Self-Assessment Questions

Answers and explanations to these questions can be found at the end of this chapter.

1. J.H. is a 30-year-old man admitted to the intensive care unit (ICU) for septic shock. He initially received 30 mL/kg of normal saline for intravenous fluid resuscitation. He required further fluid administration to improve his pulse pressure variation. Despite prophylaxis with enoxaparin 30 mg subcutaneously every 12 hours, J.H. has a proximal deep venous thrombosis. Which pharmacokinetic (PK) alteration most likely contributed to this therapeutic failure?
   A. Decreased anti-factor Xa (anti-Xa) activity secondary to decreased volume of distribution (Vd).
   B. Decreased anti-Xa activity secondary to decreased absorption.
   C. Increased anti-Xa activity secondary to decreased hepatic metabolism.
   D. Increased anti-Xa activity secondary to decreased renal elimination.

Questions 2–4 pertain to the following case.

E.W. is a 48-year-old man (height 70 inches, weight 85 kg) admitted to the trauma ICU after a motorcycle collision. E.W. presents with a traumatic brain injury (TBI; head computed tomography [CT] reveals a depressed skull fracture, frontal subarachnoid hemorrhage, and right intraparenchymal hemorrhage), right acetabulum fracture, bilateral rib fractures, and abdominal trauma. According to his abdominal CT, E.W. must go to the operating room for an exploratory laparotomy for repair of several serosal tears. After surgery, E.W. requires significant resuscitation in his first 24 hours of admission (5 L of normal saline). He is made NPO to allow bowel rest.

E.W.'s laboratory values are as follows: serum creatinine (SCr) 1.1 mg/dL, blood urea nitrogen (BUN) 17 mg/dL, and white blood cell count (WBC) 19×10^3 cells/mm^3. The patient's pulmonary artery catheterization values are cardiac index 4.2 L/minute/m^2 (normal 2.8–3.6 L/minute/m^2) and pulmonary artery wedge pressure 16 mm Hg. His medication therapy includes a fentanyl continuous infusion of 75 mcg/hour, a propofol continuous infusion of 15 mcg/kg/minute, pantoprazole 40 mg intravenously every 24 hours, enoxaparin 30 mg

subcutaneously every 12 hours, and phenytoin 150 mg intravenously every 8 hours.

2. Which most accurately assesses the risk factors for the decreased absorption of enterally administered drugs?
   A. Intestinal atrophy, pantoprazole therapy, abdominal surgery.
   B. TBI, fentanyl therapy, cardiac output.
   C. Abdominal surgery, pantoprazole therapy, TBI.
   D. Intestinal atrophy, cardiac output, fentanyl therapy.

3. Before E.W.'s ICU admission, his albumin concentration was 3.8 g/dL, but after surgery, it decreased to 2.1 g/dL. Given this change in albumin, which change in the total and unbound concentration of propofol would be most likely?
   A. Increased total concentration, decreased unbound concentration.
   B. No change in total concentration, increased unbound concentration.
   C. Increased total concentration, no change in unbound concentration.
   D. Decreased total concentration, increased unbound concentration.

4. On postoperative day 3, E.W.'s SCr increased to 3 mg/dL. On postoperative day 4, his SCr is 3.2 mg/dL. Which variable for assessing kidney function would be most important for determining E.W.'s dosing adjustments?
   A. BUN/SCr ratio.
   B. Total daily urinary output.
   C. Estimation of glomerular filtration rate (GFR).
   D. History of chronic kidney disease (CKD).

Questions 5 and 6 pertain to the following case.
S.H. is a 35-year-old man (height 70 inches, weight 85 kg) admitted to the medical ICU because of sepsis caused by hospital-acquired pneumonia. He is empirically treated with intermittent vancomycin, piperacillin/tazobactam, and ciprofloxacin. His laboratory values are as follows: SCr 1 mg/dL, BUN 12 mg/dL, and WBC 18×10^3 cells/mm^3.

5. According to S.H.'s anticipated PK changes and guideline recommendations, which would be the most appropriate intravenous loading dose of vancomycin?
   A. 1000 mg.
   B. 1500 mg.
   C. 2500 mg.
   D. 3500 mg.

6. S.H. is given a diagnosis of methicillin-resistant *Staphylococcus aureus* hospital-acquired pneumonia. On day 10 of vancomycin therapy, his vancomycin area under the curve (AUC) is 700 mg x h/L. His previous vancomycin AUC was 425 mg x h/L on the same dosing regimen. Which most likely explains what transpired?
   A. Augmented renal excretion returned to normal.
   B. Vd increased to larger than normal.
   C. Tissue penetration decreased to below normal.
   D. Liver blood flow returned to normal.

7. B.B. is 40-year-old woman with a surgical site infection caused by *Pseudomonas aeruginosa*. She is initiated on a piperacillin/tazobactam 3.375 g intravenous infusion over 4 hours every 8 hours. Which is the most likely benefit of this approach with piperacillin/tazobactam?
   A. Decreased mortality supported by prospective controlled studies.
   B. Decreased neurotoxicity supported by prospective controlled studies.
   C. Decreased mortality supported by retrospective reviews.
   D. Decreased neurotoxicity supported by retrospective reviews.

8. C.W. is a 52-year-old man admitted to the ICU for acute respiratory failure. He is given scheduled oral morphine for pain control. Which PK variable would most likely affect the hepatic metabolism of morphine?
   A. Increased α_1-acid glycoprotein (AAG).
   B. Decreased albumin.
   C. Increased hepatic blood flow.
   D. Increased intrinsic clearance.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 1: Critical Care
   a. Task A: 1, 2
   b. Task B: 2–5
2. Domain 2: Therapeutics and Patient Management
   a. Task A: 2, 3, 6
   b. Task C: 1–4
3. Domain 3: Professional Practice
   a. Task B: 1

I. INTRODUCTION

Pharmacokinetics (PK) refers to the movement of a drug throughout the body, particularly absorption, distribution, metabolism, and excretion (ADME) of a drug, whereas pharmacodynamics (PD) addresses the biochemical and physiologic effects of a drug on the body. Physiologic changes in critically ill patients cause alterations that affect the PK and PD of drugs. Although few studies evaluate the effect of these changes, clinicians must consider the general principles when making drug dosing decisions in critically ill patients. The most important consideration in critically ill patients is that changes can occur rapidly. A patient may have an altered PK variable on one day, only to experience changes that alter that variable in a completely different way on the following day. An example is a critically ill patient with augmented renal clearance (ARC) who then develops an acute kidney injury (AKI). The patient may have increased renal elimination of a specific drug, followed by decreased elimination when AKI occurs. Therefore, the critical care pharmacist should know how the principles can be altered and continually anticipate changes during a patient's stay in the ICU.

II. ROUTES OF ADMINISTRATION

A. Intravenous
   1. The intravenous route is the most widely used method of drug administration in the critically ill population. The bioavailability of an intravenously administered drug is 100%, thus ensuring the entire dose reaches the systemic circulation. As such, this route is preferred over other routes subject to reduced absorption.
   2. Although intravenous drug administration is the most popular method used in the ICU, it still poses several potential problems. The intravenous route does not guarantee penetration of the drug into sites outside the circulatory system. Examples of this include poor penetration of hydrophilic drugs into various tissues such as the meninges, pulmonary tissue, and bone. In conditions such as septic shock, drug penetration into muscle and subcutaneous tissue is lower than expected. Finally, there are reports documenting the severe adverse effects of inadvertent extravascular administration of a drug. There are several reports of drug (vesicants or non-vesicants, which can be cytotoxic or noncytotoxic) extravasation, which highlights this potential complication of intravenous administration. For example, extravasation of norepinephrine creates local vasoconstriction and tissue necrosis, which is sometimes severe enough to cause loss of a limb.

B. Enteral/Oral – Using the enteral or oral route of administration in critically ill patients results in variable drug bioavailability. The predominant concern for this route of administration in critically ill patients pertains to alterations in drug absorption. The issues pertaining to altered drug absorption are discussed in the next section. Of note, not all drugs show reduced absorption when administered enterally/orally to critically ill patients. One example comes from a study investigating the PK of atorvastatin, administered enterally. Compared with healthy volunteers, patients in the ICU had a significantly higher area under the curve (AUC) (110.5 ng/mL vs. 5.9 ng/mL, $p<0.01$) after a single dose of atorvastatin 20 mg. The increased AUC could only partly be explained by altered hepatic metabolism (Intensive Care Med 2009;35:717-21). A complicating factor with enteral administration is first-pass metabolism, which can significantly affect the bioavailability of hepatically metabolized drugs (discussed later in the chapter). As such, critical care clinicians should evaluate the use of the enteral/oral route because of both increased and decreased bioavailability with different enteral medications.

C. Subcutaneous/Intramuscular/Sublingual – Subcutaneous and intramuscular drug administration avoids first-pass metabolism by the liver and can increase the bioavailability of a drug. However, these routes still require the drug to be absorbed into the blood. Therefore, these routes are potentially affected by changes in absorption. Unlike with the enteral or oral route, clinicians do not routinely abandon the use of subcutaneously or intramuscularly administered medications. Examples include the continued use of low-molecular-weight heparins, insulin, and the antipsychotic haloperidol in patients with obesity for whom the absorption may be altered. In addition, clinicians should be aware of similar concerns for altered absorption in critically ill patients with sepsis or shock states because of changes in perfusion. Use of the sublingual route of administration in critically ill patients has not been studied. However, one could expect that alterations in blood flow and absorption would cause similar concerns as for subcutaneous and intramuscular routes. Critical care clinicians should understand that the subcutaneous route may be used; however, little to no data are available in critically ill patients to determine if any pharmacokinetic changes will occur.

D. Inhalation – Administration of drugs directly into the lungs of critically ill patients is usually chosen to reduce systemic exposure and/or achieve a high concentration in the pulmonary tissue. The high local concentration is intended to maximize the therapeutic effect while reducing any adverse or unwanted effects. For example, the use of inhaled bronchodilators reduces unwanted systemic effects such as tachycardia. Antibiotics such as colistin and aminoglycosides are administered via inhalation to improve the antibiotic concentrations in the lungs and reduce exposure to the kidneys. Studies have demonstrated varying serum concentrations following inhaled administration of amikacin in critically ill patients, but these concentrations were much lower than when administered intravenously (J Antimicrob Chemotherapy 2016;71:3482-6). Additional research found that older age and higher positive end expiratory pressure (PEEP) independently predicted detectable serum tobramycin concentrations (Respir Care 2022;67:16-23). While inhaled antibiotics are used to reduce the likelihood of systemic adverse effects, their efficacy of this route of antibiotic administration is not well established. Drug particles of 1–5 micrometers have the best opportunity to be delivered to all areas of the lungs. Smaller particles will be exhaled without being deposited in the lower airways, whereas larger particles will be deposited in the large bronchi or the oropharynx. Several models of nebulizers are on the market that use different methods to achieve the desired particle sizes.

E. Intrathecal/Intraventricular – Use of the intraventricular route is intended to increase local concentrations and reduce systemic concentrations. Data evaluating the efficacy of this route of administration are lacking in the general population limited to case series. Despite the lack of data, clinicians use this route when treating multidrug-resistant meningitis with antibiotics, or for the administration of analgesics in patients with chronic pain.

Patient Case

1. M.J. is a 70-year-old man admitted to the neurosurgical ICU for an aneurysmal subarachnoid hemorrhage. His initial treatment included placement of an external ventricular drain. Subsequently, he had a maximum temperature of 101.5°F, a WBC of 15×10^3 cells/mm^3, and a cerebrospinal fluid culture positive for methicillin-resistant *S. aureus*. Intraventricular vancomycin 20 mg is used for therapy. Which is the best rationale for this approach?
   A. Demonstrated superiority to intravenous antibiotics.
   B. Maximizing localized antibiotic concentrations.
   C. Safer administration method.
   D. Reducing the ototoxicity of vancomycin.

III. ABSORPTION

A. Bioavailability (F) refers to the percentage of an administered dose of drug that reaches the systemic circulation. Bioavailability from subcutaneous and intramuscular or enteral administration is affected by absorption and first-pass metabolism (enterally administered drugs). Few studies directly assess the enteral absorption of drugs in critically ill patients, and the results vary. In addition, studies of enterally administered drugs do not differentiate whether plasma concentrations are altered because of changes in absorption or first-pass metabolism. Although data on absorption in critically ill patients are limited, clinicians must consider several factors if a route of administration other than intravenous is preferred.

B. Gastrointestinal (GI) Perfusion – Hypotension and/or shock are known to cause the shunting of blood toward the vital organs (e.g., brain, heart, lungs) and away from the less vital organs (e.g., muscles, skin, splanchnic organs), which can alter drug absorption.
   1. GI absorption: No studies clearly show the effect of hypotension or shock on the oral or enteral absorption of drugs. Clinicians extrapolate changes in splanchnic blood flow to the likelihood that GI absorption is altered. Redistribution of blood away from the splanchnic circulation is thought to decrease drug absorption from the GI tract. The hyperdynamic phase of sepsis or septic shock can increase cardiac output, and studies have shown an increase in hepatosplanchnic (portal vein and hepatic artery) blood flow. In late-stage (decompensated) sepsis, it is thought that splanchnic blood flow is decreased, but no studies have verified this. This uncertainty in splanchnic blood flow and GI absorption leads many clinicians to forgo the enteral route for drug administration. One diagnostic test to consider to evaluate GI absorption is an oral acetaminophen test. Of note, one study evaluated the use of acetaminophen at a dose of 15 mg/kg with serum concentrations within 3 hours of administration. The majority of patients had detectable serum acetaminophen concentrations between 30 and 60 minutes with no adverse outcomes associated with its use, suggesting this intervention as a potential surrogate for GI absorption (Am J Ther 2023;30:e95-102).
   2. Transdermal, subcutaneous, and intramuscular absorption: No studies have evaluated the effect of hypotension or shock on transdermal, subcutaneous, or intramuscular absorption. Similar to splanchnic circulation, the shunting of blood to vital organs reduces blood flow to the skin and muscles, which is thought to reduce absorption from these sites. This assumption is supported by the observation that critically injured trauma patients with edema have significantly lower anti-Xa and antithrombin activity after treatment with subcutaneous enoxaparin (Ann Surg 2023;277:734-1). Altered absorption is believed to be a contributing factor to these results. However, this may not be generalized to all critically ill patients because anti-Xa activity was not significantly different in edematous compared with nonedematous medical-surgical ICU patients after dalteparin administration (Crit Care 2006;10:R93). Furthermore, the direct effects of critical illness on coagulation may alter the pharmacodynamics of low molecular weight heparins and obscure the degree to which absorption impacts their effect.
   3. Vasopressor effect: Vasopressors may contribute to regional hypoperfusion, which could result in decreased drug absorption. Vasopressin reduces splanchnic blood flow in patients with distributive shock. In septic shock, epinephrine results in reduced splanchnic blood flow. Dopamine is not as effective as norepinephrine in maintaining splanchnic blood flow in patients with stable distributive shock. Conversely, when gut perfusion is compared between cardiac surgery patients with and without vasodilatory shock, norepinephrine use results in higher intestinal perfusion. However, this is countered by a worse splanchnic oxygen demand versus supply. The variable effect of vasopressors on splanchnic perfusion creates enough concern that most clinicians abandon the use of orally or enterally administered drugs when vasopressors are being used. The large inter-individual pharmacokinetic variability seen with enterally administered fludrocortisone in patients with septic shock illustrate this concern. In addition, absorption from other sites could be impaired. One study investigated the anti-Xa

activity of the low-molecular-weight heparin certoparin in critically ill patients, of whom 40.3% were receiving vasopressors. Less than 50% of patients receiving standard doses of certoparin had anti-Xa activity in the antithrombotic range (0.1–0.3 IU/mL) (Crit Care 2005;9:R541-8).

C. Intestinal Atrophy – After 3–5 days of fasting, gut mucosal crypt depth and villus height can be decreased. This correlates with an abnormal lactulose-mannitol test, indicating increased gut permeability. Splanchnic hypoperfusion can further worsen gut hypoxia, exacerbating gut permeability. However, the effect of intestinal atrophy on drug absorption has not been systematically evaluated. Atrophy and the corresponding loss of integrity of the tight junctions could lead to an increased absorption of drugs that are absorbed through passive diffusion. Conversely, cellular dysfunction caused by atrophy might decrease the absorption of drugs that require active transport for absorption. Currently, no studies can clarify this quandary.

D. GI dysmotility – GI dysmotility has clearly been established in critically ill patients, with an incidence as high as 60%. Box 1 shows the conditions in critically ill patients that are associated with delayed gastric emptying caused by dysmotility. Acetaminophen kinetics show that GI dysmotility causes a delay in absorption and a reduced peak concentration in most studies. Concern regarding PK changes in the presence of delayed gastric emptying is a major factor contributing to the avoidance of orally or enterally administered drugs in critically ill patients. GI dysmotility is usually treated using prokinetic agents such as metoclopramide or erythromycin. No data exist regarding the effect of prokinetics on drug absorption in critically ill patients with dysmotility. However, one study found that, in healthy individuals, coadministration of erythromycin and the controlled-release formulation of pregabalin resulted in a 17% decrease in AUC and a 13% decrease in peak plasma concentrations of pregabalin (Clin Drug Investig 2015;35:299-305). This suggests that prokinetics can affect the absorption of other medications. In the critically ill patient, the effect of prokinetics on the PK of orally or enterally administered remains relatively unclear.

Box 1. Reasons for Delayed Gastric Emptying

Burns	Mechanical ventilation	Surgery
Electrolyte abnormalities	Opioid analgesics	Trauma
GLP-1 agonists	Postoperative ileus	Traumatic brain injury
Hyperglycemia	Sepsis	
Ileus	Shock	

E. Intestinal Drug Transporters – Transmembrane proteins such as P-glycoprotein (P-gp) and cytochrome P450 (CYP) enzymes play an integral role in drug absorption. In general, these transporters reduce the absorption of drug substrates. Therefore, decreased activity of these enzymes will theoretically increase the absorption of drugs that are substrates. Conversely, several intestinal transporters facilitate drug absorption and may thus decrease drug absorption. However, no PK studies evaluate changes in drug absorption caused by changes in intestinal transporters that are specifically related to critical illness or conditions often present in these patients. Increased proinflammatory cytokines in patients with conditions such as systemic inflammatory response syndrome and sepsis affect P-gp expression and activity. Therefore, enteral drug absorption can be altered in these states. However, no studies have directly investigated the effects of inflammatory states, or systemic inflammatory response syndrome, on drug absorption changes mediated by changes in P-gp activity.

F. Physical Incompatibilities – Drugs administered through enteral feeding tubes come in contact with gastric secretions, intestinal secretions, and enteral nutrition formulas, all of which may pose a problem for drug absorption.
   1. Drug-enteral-nutrition interactions: Some drugs potentially interact with enteral nutrition. The degree of interaction and clinical significance varies.

a. Ciprofloxacin bioavailability is reduced when it is administered with enteral nutrition (with the exception of the oral suspension), but most studies suggest that serum concentrations remain above the minimum inhibitory concentration (MIC) for most bacterial pathogens.
b. Enteral nutrition has been reported to significantly reduce the absorption of levothyroxine, phenytoin, and warfarin. One case report showed a reduction in voriconazole serum concentrations when enteral nutrition was initiated (J Oncol Pharm Pract 2012;18:128-31).
c. A suggested solution to this interaction is to hold the enteral nutrition 1–2 hours before and after drug administration. However, this poses two problems. First, interruption of enteral nutrition may contribute to inadequate nutrition support. Second, this increases the difficulty of administering the medications appropriately. Increasing the workload of nurses can lead to administration delays or, even worse, errors. This is very important, because failure to withhold the enteral nutrition could result in suboptimal dosing and effects of the interacting drug.

2. pH changes: The state of ionization of a drug usually affects the lipophilicity and potentially the absorption. Examples from non-critically ill patients include increased gastric pH caused by histamine-2 receptor antagonists or proton pump inhibitors (both commonly used in the critically ill population for GI prophylaxis), resulting in decreased absorption of ketoconazole, itraconazole, atazanavir, indinavir, dasatinib, mycophenolate mofetil, cefpodoxime, and dipyridamole. Acid-suppressive drugs increased nifedipine and digoxin absorption, and alendronate had a 2-fold increase in bioavailability in the presence of these agents (Aliment Pharmacol Ther 2009;29:1219-29). Additionally, drug coating or extended release matrices may be altered by pH changes and could potentially impact oral absorption. For example, coadministration of a proton pump inhibitor with enteric coated ketoprofen altered the maximum concentration and time to maximum concentration. Drug absorption depends on the drug's solubility and ability to permeate the intestinal mucosa. The distal end of the feeding tube can be in the stomach, duodenum, or jejunum. Many drugs must be administered into the stomach or duodenum so that they can be properly dissolved by gastric juices, bile, and pancreatic enzymes and can be fully absorbed through the intestines. Thus, drugs like warfarin, which is absorbed high up in the small bowel, and oral iron, which is dissolved in the stomach and absorbed in the duodenum, might not be properly absorbed if they are administered via a jejunostomy tube.

G. Important Considerations for Absorption

1. The overall uncertainty of a patient's ability to absorb drugs from the GI tract often results in the clinician's avoidance of enterally administered drugs because they want to ensure adequate bioavailability. The decision to use the GI route of administration is arbitrary, and in some cases this approach may be more costly. Many clinicians anecdotally use tolerance of enteral feedings as a surrogate marker for normal drug absorption. The definition of tolerance is debatable and assessment of tolerance is highly variable, so it is not an ideal maker for drug absorption. If a drug known to interact with enteral nutrition requires enteral administration, nutrition should be interrupted for the minimum duration possible around drug administration.
2. Subcutaneous and intramuscular routes of administration present similar problems with absorption, but clinical practice has not abandoned these routes of administration. Some clinicians advocate using larger doses of drugs being administered subcutaneously, but no studies have verified the safety or efficacy of this practice. Small studies investigating enoxaparin dosing in medically ill and trauma patients with obesity found improved anti-Xa serum concentrations when using a weight-based dosing method (0.5 mg/kg every 12 hours) compared with standard dosing. This suggests that larger subcutaneous fat distribution can alter the pharmacokinetics and PD of some medications (Thromb Res 2010;125:220-3; Am J Surg 2013;206:847-51).

Patient Case

2. I.L. is a 32-year-old man receiving stress ulcer prophylaxis with esomeprazole 40 mg intravenously every day. Which drug will most likely have an increased absorption secondary to the increased gastric pH?
   A. Carvedilol.
   B. Ciprofloxacin.
   C. Diazepam.
   D. Digoxin.

IV. DISTRIBUTION

A. The volume of distribution (Vd) of a drug is a PK variable that relates the total amount of drug in the body to the concentration of drug in the serum. A simple mathematical representation of this relationship is the following equation:

$$C = \frac{\text{dose}}{\text{Vd}}$$

where *C* is the initial serum concentration of an intravenously administered drug and Vd is the volume of distribution. However, the distribution of most drugs is more complex and is affected by several factors such as systemic and tissue perfusion, degree of protein binding, tissue permeability, drug lipid solubility, drug pKa, and pH of the environment. Critically ill patients may be subjected to one or more changes in the previously stated factors that could result in an altered Vd for some drugs.

B. Tissue Perfusion – As noted in the previous section, shock states and vasopressor use can cause a redistribution of blood flow. This results in decreased perfusion of the muscle, skin, and splanchnic organs. Hydrophilic drugs with a smaller Vd (ones that remain in the plasma water volume) may have decreased distribution to physiological compartments with decreased blood flow. This is highlighted by animal studies of septic shock showing lower gentamicin concentrations in the microcirculation than in the central vessels.

C. Fluid Shifts and Tissue Membrane Permeability – Critically ill patients can receive significant volumes of intravenous fluid for resuscitation. This often results in increased total body water and interstitial volume. In addition to fluid administration, diseases such as sepsis, thermal injury, acute respiratory distress syndrome, AKI, heart failure, and cirrhosis can increase interstitial fluid volumes. Moreover, surgery increases extracellular volume postoperatively. In this setting, the Vd for hydrophilic drugs is increased, whereas the Vd for lipophilic drugs is often unchanged. The increased interstitial water provides a larger compartment for hydrophilic drugs to distribute, thus decreasing the serum concentrations. In addition, because distribution is into a larger interstitial space (as the result of increased interstitial water), the drug concentration can be decreased in this space. This has been shown in microdialysis studies evaluating subcutaneous tissue concentrations for intravenously administered piperacillin. Compared with healthy volunteers, patients with septic shock had reduced piperacillin tissue concentrations. However, the increased Vd of drugs is not universally noted with increased interstitial fluid volumes in some critically ill patients. Although one study found increases in aminoglycoside Vd, another study was unable to correlate fluid shifts with changes in the aminoglycoside Vd in critically ill surgical patients (Crit Care Med 1988;16:327-30).

D. Protein Binding – Many drugs are bound to serum protein carriers such as albumin, AAG, lipoproteins, and cortisol-binding protein. Albumin and AAG are particularly important in critically ill patients because albumin usually binds to acidic drugs (e.g., diazepam, phenytoin), whereas AAG binds to basic drugs (e.g., lidocaine, diltiazem). Of importance, albumin concentrations usually decrease (negative acute phase reactant) under stress, while AAG concentrations increase (positive acute phase reactant). The following equation represents the calculation of Vd:

$$Vd = \left(\frac{f_u}{f_{ut}} \right) Vt + Vp$$

where f_u is the fraction unbound in the plasma, f_{ut} is the fraction unbound in the tissues, Vt is the volume of tissue, and Vp is the volume of plasma. When the plasma concentration of albumin decreases, the f_u of a drug increases. This increase results in an increased Vd. Conversely, increases in AAG plasma concentrations decrease the f_u of a drug bound to AAG due to increased protein binding, ultimately decreasing the Vd for that drug.

1. The clinical relevance of this was noted when a decrease in the Vd of lidocaine correlated with an increase in AAG in post-cardiac surgery patients. It was suspected that arrhythmias were caused by these PK changes (Clin Pharmacol Ther 1984;35:617-26). A similar effect has been reported where development of hypoalbuminemia was associated with elevated free valproate concentrations (Crit Care Explor 2022;4:e0746).
2. Table 1 provides examples of drugs used in critically ill patients that bind to albumin and AAG.

Table 1. Extraction Ratio (ER) and Protein Binding of Select Drugs Used in Critically Ill Patients

Protein Binding	High ER Drugs[a]	Intermediate ER Drugs[a]	Low ER Drugs[a]
Albumin	Diltiazem Morphine Propofol Propranolol Verapamil	Aspirin Carvedilol Midazolam Omeprazole	Carbamazepine Ceftriaxone Dexamethasone Diazepam Itraconazole Phenytoin Valproic acid Warfarin
AAG	Diltiazem Fentanyl Lidocaine Propranolol Verapamil	Midazolam	Carbamazepine

[a]ER is addressed in section V. Metabolism.

AAG = α_1-acid glycoprotein.

E. pH – Acid-base disorders are common among the critically ill population. Although these disorders are treatable, they create plasma pH changes that could affect drug distribution. Most drugs are either weak acids or bases and exist in either the ionized or the nonionized state, depending on the surrounding environment. Nonionized drugs penetrate cell membranes more easily than do ionized drugs. Therefore, a drug in the ionized state would be expected to have a smaller Vd than when in the nonionized state. Theoretically, a drug that is a weak acid in a patient experiencing acidemia would be expected to have a larger Vd, whereas a basic drug would have a smaller Vd in the same patient. Although the potential exists to correlate plasma pH changes with changes in drug Vd, evidence in humans is lacking.

F. Extracorporeal Membrane Oxygenation – Use of extracorporeal membrane oxygenation (ECMO) in critically ill patients has increased over the years. Depending on the institution and clinician expertise, ECMO can play a significant role in the care of critically ill patients with respiratory and/or circulatory failure. With implementation of this therapy, changes in the PK of medications have been described. The most common alteration reported is an increase in Vd. Use of ECMO therapy consists of a large volume of blood being extracted from the patient through catheter tubing (generally polyvinyl chloride), circulated through an oxygenator and a heat exchanger before the newly oxygenated blood is returned to the patient. Depending on the type of circuit and/or pump used for ECMO, the volume required to prime the circuit can increase the Vd of some commonly used medications, especially those with a small Vd. Important factors to consider when evaluating the PK changes in medications when ECMO is implemented include (1) data evaluating changes are limited and (2) most data analyses available are in neonates and pediatric patients; thus, the published data will not always translate to the same effects in adults. Underdeveloped kidney and liver function in neonates contribute to altered PK that may not be present in critically ill adults. Adult data analyses are limited to mostly case reports and retrospective data. Prospective studies in adults include evaluations of oseltamivir, vancomycin, and meropenem (AACN Adv Crit Care 2018;29:246-58). According to the prospective data, the degree of change in PK created by ECMO may be less than that found in retrospective studies, at least for the studied medications. The generally anticipated pharmacokinetic changes created by ECMO are the result of three factors: (1) the ECMO circuit tubing and membrane oxygenator may bind medications, causing drug sequestration, and the resulting PK change expected is an increased Vd; (2) circuit priming fluid type, fluid pH, and volume potentially increase in medication Vd; and (3) as the ECMO circuit ages, medication binding becomes saturated, creating a scenario in which patient medication requirements may return back to pre-ECMO dosing. Several ex vivo studies have demonstrated loss of drug in the ECMO circuit, including fentanyl and midazolam. This effect should prompt clinicians to monitor patients closely for proper analgesia and sedation during ECMO treatment (Crit Care 2015;20:40., Intensive Care Med 2007;33:1018-1024). Other PK changes expected with ECMO relate to the critically ill state of the patient and are discussed throughout this chapter. Of note, these concepts are generalizations, and data for specific drugs may differ. For example, an ex vivo study investigating drug binding to ECMO circuits found that ciprofloxacin recovery rates were 96%, even though the drug was lipophilic and expected to bind to circuit tubing (Crit Care 2015;19:164). A prospective, observational, pilot study evaluated β-lactam and aminoglycoside PK parameters in patients receiving ECMO. Study results showed variable achievement of PK objectives, which led the authors to recommend therapeutic drug monitoring for patients receiving ECMO concurrently with β-lactam or aminoglycoside antibiotics (Anaesth Crit Care Pain Med 2019;38:493-97). Physiochemical properties of the drug molecule also determine the interaction with the circuit. Compound characteristics that should be considered include pKa and degree of ionization, molecular size, plasma protein binding, and lipophilicity. Drug lipophilicity is often reported through the n-octanol/water partition coefficient, or logP, whereas a high logP indicates a more lipophilic compound at higher risk of ECMO circuit sequestration. Protein binding and the extent thereof may also influence the extent of drug sequestration in the ECMO circuit (Crit Care 2015;19:164). The COVID-19 pandemic often necessitated the use of ECMO and revealed pharmacokinetic changes in several antiviral agents. Changes in Vd were found for favipiravir and ribavirin. Suspected alterations in cytochrome P450 metabolism were noted for chloroquine, lopinavir, ritonavir, and favipiravir. The reader is referred to a comprehensive review by the European Society of Clinical Microbiology, Infectious Diseases (Clin Pharmacokinet 2020;59:1195-16) for more information about the impact of COVID-19 and ECMO on the pharmacokinetics of antivirals.

Patient Cases

3. R.H. is 20-year-old man who presents to the emergency department with nausea and vomiting. His vital signs are significant for heart rate 130 beats/minute, blood pressure 98/62 mm Hg, and respiratory rate 28 breaths/minute. Laboratory tests reveal an arterial blood gas significant for a pH of 7.11, Pco_2 of 18 mm Hg, and bicarbonate of 5.2 mEq/L. His basic metabolic panel is significant for potassium concentration 5 mEq/L, BUN 22 mg/dL, SCr 1.4 mg/dL, and blood glucose 400 mg/dL. Which changes would be most likely to occur in the Vd of a weak acid like ciprofloxacin in this patient?
   A. Increased because of decreased ionization.
   B. Decreased because of increased ionization.
   C. No change because of no change in ionization.
   D. Decreased because of decreased ionization.

4. B.B. is a 62-year-old woman admitted to the ICU for septic shock. She has received 6 L of crystalloids in the past 24 hours, including her initial resuscitation and maintenance fluids. Which of the following antibiotic volumes of distribution would most likely be unaffected by the amount of intravenous fluids administered?
   A. Tobramycin.
   B. Daptomycin.
   C. Levofloxacin.
   D. Cefepime.

V. METABOLISM

A. Introduction – The predominant location for drug metabolism is the liver, but it can include tissues such as the GI tract, kidneys, lung, and brain. The greatest extent of knowledge regarding drug metabolism and, more importantly, changes in critically ill patients relates to hepatic metabolism. Therefore, this section will focus largely on changes in the hepatic metabolism of drugs, specifically high and low extraction ratio drugs. However, the next section briefly discusses renal metabolism because the clinical ramifications of drug metabolism in the kidney are a potential area for future research.

B. Renal Metabolism – There is evidence that the kidneys express the CYP isoenzymes 2B6 and 3A5. Data suggest that CYP 2C8, 2C9, and 3A4 are also expressed in the kidneys. In addition, UGT (UDP-glucuronosyltransferase) enzymes 1A9 and 2B7 are abundantly expressed in the kidneys and play a role in the glucuronidation of drugs. However, no data describe how changes in critically ill patients affect drug metabolism in the kidneys by these enzymes. As evidenced by increased hypoglycemic events and lower insulin requirements in critically ill patients with AKI, clinically relevant changes in renal insulin metabolism occur, but the exact mechanisms for these are not well characterized (Nutrition 2011;27:766-72).

C. Hepatic Metabolism – Hepatic clearance refers to the volume of blood that is completely cleared of drug by the liver per unit of time. The ability of the liver to metabolize drugs depends on three physiologic variables: hepatic blood flow, drug protein binding, and the intrinsic activity of hepatic enzymes. When evaluating an intravenously administered drug (bioavailability of 1), clearance by the liver can be simply represented by the following equation:

$$CL_H = Q \times E$$

where CL_H is the hepatic clearance, Q is the hepatic blood flow, and E is the hepatic extraction ratio. The hepatic extraction ratio can further be described by the following equation:

$$E = \frac{f_u \times CL_{int}}{Q + f_u \times CL_{int}}$$

where f_u is the fraction unbound in the plasma, CL_{int} is the intrinsic hepatic clearance, and Q is hepatic blood flow. The hepatic extraction ratio is classified by the fraction of drug removed during one pass through the liver and can range from 0 to 1. It can be separated into high (greater than 0.7), intermediate (0.3–0.7), and low (less than 0.3) categories. The extraction ratio would be zero when the liver does not metabolize a drug and 1 when CL_H depends entirely on hepatic blood flow. The effect of changes in critical illness depends on the extraction ratio of the drug. Table 1 lists select high extraction ratio and low extraction ratio drugs.

D. High Extraction Ratio Drugs

1. Drugs with a high hepatic extraction ratio are metabolized by hepatic enzymes and are thus cleared by the liver. In drugs with high extraction ratios, clearance does not vary with changes in hepatic enzymatic activity and is primarily limited by hepatic blood flow. Mathematically, this can be represented by:

$$f_u \times CL_{int} >>> Q$$

Given the previously stated relationship, CL_H can be simplified to:

$$CL_H = Q$$

As previously noted, clearance of a drug pertains to removal of the drug from the blood. Therefore, the effect on plasma drug concentration will affect the efficacy of the drug. Because an unbound or "free drug" elicits the pharmacological response, the unbound steady-state concentration (C_{ssu}) is important to assess. The total concentration, otherwise called the steady-state concentration (C_{ss}) of hepatically metabolized drugs, can be represented by the following equation:

$$C_{ss} = \frac{dose}{CL_H}$$

where C_{ss} is the steady-state concentration (for both protein bound and unbound drugs) and dose is the rate of drug input. Because $CL_H = Q$ for high extraction ratio drugs, the equation can be modified to:

$$C_{ss} = \frac{dose}{Q}$$

The unbound steady-state concentration for a high extraction ratio is represented by the following equation:

$$C_{ssu} = \frac{f_u \times dose}{Q}$$

Figure 1 shows how a change in each variable of CL_H affects the C_{ss} and C_{ssu}. For high extraction ratio drugs, altered hepatic blood flow affects both the C_{ss} and the C_{ssu}, whereas changes in f_u affect only the C_{ssu}.

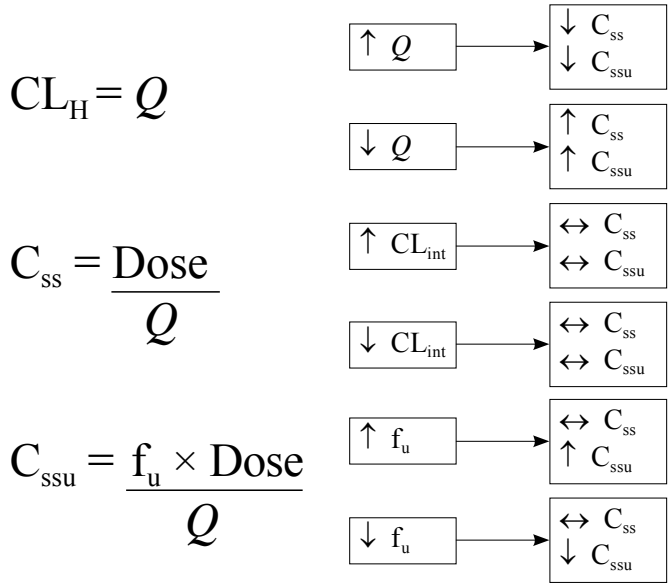


Figure 1. Effect of variable changes on steady-state and unbound steady-state concentrations of a high extraction ratio drug.

2. Effect of increased hepatic blood flow: Animal models have shown a clear increase in splanchnic perfusion during the hyperdynamic phase of sepsis. Critically ill patients in the hyperdynamic phase of sepsis or septic shock have an increased cardiac output and increased hepatosplanchnic blood flow. However, no correlation directly relating cardiac output (or an increase in cardiac output) with an increase in splanchnic blood flow could be established in some studies. Therefore, quantification of the increase in blood flow and the resultant increase in hepatic metabolism cannot be established. The clinician is left to assume the potential for increased metabolism of high extraction ratio drugs and the expected decrease in unbound steady-state concentration and possibly a reduced clinical efficacy.
    a. Any condition that increases cardiac output could theoretically improve hepatic blood flow. Although data in humans are lacking, an animal model of endotoxemia showed improved hepatic blood flow after dobutamine.
3. Effect of decreased hepatic blood flow
    a. Conditions with a low cardiac output such as hypovolemic or hemorrhagic shock, decompensated sepsis, myocardial infarction with or without cardiogenic shock, and acute heart failure exacerbation would be expected to cause a decrease in hepatic blood flow. Human studies to verify this assertion are lacking. Animal models of decompensated sepsis and cardiogenic shock show reduced hepatic blood flow, which would be expected to increase the C_{ssu} and potentially increase the effect of, or produce toxicity for, drugs with high extraction ratios.
    b. Mechanical ventilation may also decrease cardiac output and subsequently hepatic blood flow because of increased intrathoracic pressure. This pressure causes a decrease in venous return to the heart, compresses the ventricles, and reduces ventricular filling. The result is a decrease in cardiac output (N Engl J Med 1981;304:387-92) and hepatic blood flow (Crit Care Med 1982;10:703-5).

E. Low Extraction Ratio Drugs – Drugs with a low hepatic extraction ratio undergo a lower degree of hepatic enzyme metabolism; therefore, clearance is limited by hepatic enzymatic activity, and clearance is independent of hepatic blood flow. Mathematically, this can be represented by:

$$f_u \times CL_{int} <<< Q$$

According to this relationship, CL_H can be simplified to:

$$CL_H = f_u \times CL_{int}$$

Again, the C_{ss} for hepatically metabolized drugs can be represented by the following equation:

$$C_{ss} = \frac{dose}{CL_H}$$

where C_{ss} is the steady-state concentration and dose is the rate of drug input. Because $CL_H = f_u \times CL_{int}$ for low extraction ratio drugs, the equation can be modified to:

$$C_{ss} = \frac{dose}{f_u \times CL_{int}}$$

The unbound steady-state concentration for a low extraction ratio drug is represented by the following equation:

$$C_{ssu} = \frac{dose}{CL_{int}}$$

Figure 2 shows how a change in each variable of CL_H affects the C_{ss} and the C_{ssu}. For low extraction ratio drugs, altered CL_{int} affects both the C_{ss} and the C_{ssu}, whereas changes in the f_u affect only the C_{ss}.

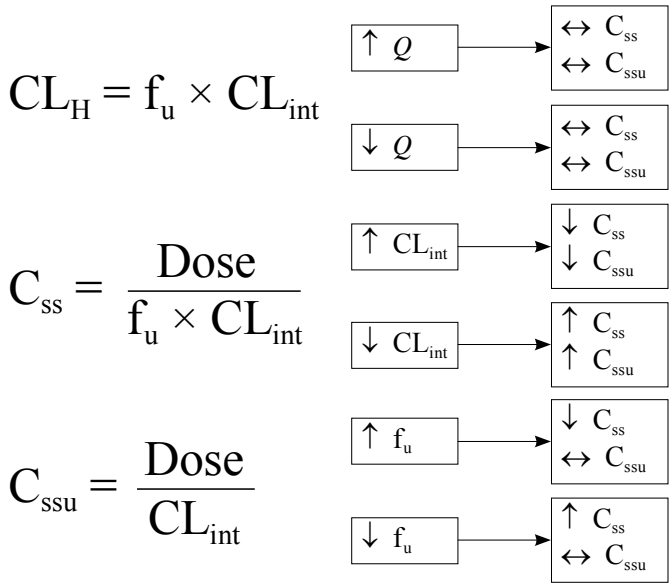


Figure 2. Effect of variable changes on steady-state and unbound steady-state concentrations of a low extraction ratio drug.

F. Effect of Changes in Intrinsic Clearance
   1. Drug interactions – A major mechanism for altered intrinsic clearance is not caused by changes in critical illness, but still poses a significant threat to altered metabolism of drugs. The CYP enzymes play an important role in phase I metabolism. Many drugs used in critically ill patients are substrates, inducers, inhibitors, or combinations of these. Critically ill patients often have complex pharmacotherapeutic regimens that create the potential for drug interactions through the CYP system. As in other patient populations, drug concentrations are increased when substrates are coadministered with inhibitors of the same CYP and decreased when substrates are coadministered with inducers.
   2. Inflammation – Inflammatory states play an important role in altering CYP activity. The inflammatory cytokines interleukin (IL)-1, IL-6, and tumor necrosis factor alpha decrease the expression and activity of CYP enzymes. Similarly, patients in early sepsis would have increased inflammatory cytokines with the resultant depressed CYP activity. This is supported by studies showing that endotoxin administration results in decreased CYP-mediated drug metabolism in healthy volunteers. However, studies have not characterized a time course for the cytokine-mediated changes. Clinicians are left to use patient response and monitoring for toxicity to determine whether drug metabolism is altered or has returned to normal.
   3. Hypothermia – Animal models have shown that hypothermia affects drugs metabolized through the CYP system. Drugs studied in animal models include fentanyl, pentobarbital, propranolol, and morphine. Human studies have investigated the effect of hypothermia on low extraction ratio drug kinetics. One example showed changes in phenytoin PK during mild hypothermia. Specifically, increased concentrations and reduced metabolism, but no changes in protein binding, were noted during hypothermia (Ther Drug Monit 2001;23:192-7). Other drugs noted to have decreased hepatic clearance during hypothermia are midazolam, fentanyl, remifentanil, phenobarbital, and vecuronium.
   4. AKI – One study investigated the effects of AKI on the hepatic metabolism of midazolam. Patients with worsening AKI, as determined using the RIFLE (risk, injury, failure, loss, end-stage kidney disease) criteria, had increasing midazolam concentrations. The authors hypothesized that the increased concentrations were caused by impaired CYP3A activity (Intensive Care Med 2012;38:76-84).

G. Intermediate Extraction Ratio Drugs – Metabolism of an intermediate extraction ratio drug depends on hepatic blood flow, intrinsic clearance, and fraction of unbound drug. Essentially, intermediate extraction ratio drugs depend on the same variables as both the low extraction ratio and high extraction ratio drugs. As such, they are the most complex drugs for determining how hepatic clearance will be affected in critically ill patients. This is important because critically ill patients usually have more than one change occurring at the same time. For example, patients with septic shock may have increased hepatic blood flow secondary to increased cardiac output while having a decreased intrinsic clearance secondary to increased circulating inflammatory cytokines. Quantifying the overall effect is difficult in the ever-changing critically ill patient. The clinician is often left to monitor for the expected therapeutic outcome while being aware of the potential toxicities. Table 1 lists select intermediate extraction ratio drugs.

H. Other Factors
   1. TBI increases the hepatic clearance of some drugs.
      a. One study found that patients with TBI had increased hepatic clearance of phenytoin during the first 7–14 days. The authors alluded to the possibility that the increased clearance was associated with changes in protein binding, induction of metabolism, or stress on hepatic metabolic capacity (Clin Pharmacol Ther 1988;44:675-83).
      b. Another study noted a correlation between nutritional protein intake and increased phenytoin metabolism in patients with TBI.

c. Phase II enzymatic activity may also be enhanced in patients with TBI, as evidenced by increased lorazepam clearance. Similar data for lorazepam were noted in thermally injured patients. These data suggest that phase II metabolism can be affected by critical illness.

2. Hepatic failure – Hepatic failure will significantly affect drug dosing in the critically ill patient. See the Hepatic Failure/GI/Endocrine Emergencies chapter for more information regarding drug dosing in hepatic failure.

Patient Cases

5. A.P. is a 35-year-old woman admitted to the ICU for an acute asthma exacerbation. She was intubated and required mechanical ventilation. She is prescribed morphine for pain control. Which best describes the effect of mechanical ventilation on morphine unbound concentrations?

   A. Increases oxygenation delivery to the liver and increases the unbound concentration.
   B. Decreases hepatic blood flow and increases the unbound concentration.
   C. Increases cytokine production and decreases the unbound concentration.
   D. Cannot affect the unbound concentration.

6. J.M. is a 34-year-old woman admitted to the ICU with new-onset sepsis. She was continued on her home medication, diazepam, for anxiety. She is empirically initiated on vancomycin 1 g intravenously every 12 hours, piperacillin/tazobactam 3.375 g intravenously every 6 hours, and fluconazole 400 mg intravenously every 24 hours. Which PD response would you most expect in J.M.?

   E. Increased risk of anxiety caused by an increased intrinsic clearance of diazepam.
   F. Increased risk of sedation caused by a decreased intrinsic clearance of diazepam.
   G. Increased risk of anxiety caused by a decreased unbound fraction of diazepam.
   H. Increased risk of sedation caused by an increased unbound fraction of diazepam.

7. P.M. is receiving phenytoin for the treatment of posttraumatic seizures. The measured total phenytoin concentration is 8 mcg/mL. You calculate the adjusted concentration according to P.M.'s hypoalbuminemia (albumin of 2.5 g/dL) and determine the concentration to be approximately 13 mcg/mL. Which of the following relationships best describes why an adjusted total phenytoin concentration must be calculated?

   A. The phenytoin C_{ssu} is increased because of an increase in the f_u.
   B. The phenytoin C_{ss} is increased because of an increase in f_u.
   C. The phenytoin C_{ssu} is decreased because of an increase in f_u.
   D. The phenytoin C_{ss} is decreased because of an increase in f_u.

8. C.P. is a 50-year-old man admitted to the medical ICU for management of diabetic ketoacidosis. His medical history is significant for hypertension, type 1 diabetes, and a myocardial infarction 2 years ago. He quit smoking last year and drinks alcohol occasionally. His vital signs are significant for heart rate 125 beats/minute, blood pressure 95/65 mm Hg, and respiratory rate 22 breaths/minute. His significant hypovolemia contributed to the development of AKI. His current SCr is 2.8 mg/dL. His blood glucose is significantly elevated at 350 mg/dL. He will be initiated on a continuous intravenous infusion of insulin to correct his blood glucose. Which factor is most important to consider when dosing insulin in C.P.?

   A. Decreased renal metabolism of insulin.
   B. Increased Vd of insulin.
   C. Increased hepatic metabolism of insulin.
   D. Decreased receptor binding of insulin.

VI. EXCRETION

A. Renal Excretion
1. For most drugs, the kidneys are the primary site for excretion of the parent drug, metabolites, or both. Urinary excretion of a drug depends on filtration, secretion, and reabsorption. Patients in the ICU may have increased, decreased, or normal renal excretion of drugs. The state of renal excretion depends on many variables and can change rapidly. This is especially true in ICU patients, for whom a clinical condition can contribute to both increased and decreased excretion, depending on how that condition progresses.
2. Filtration
   a. Glomerular filtration rate (GFR) is the variable most widely used to describe kidney function. The National Kidney Foundation defines normal kidney function as 140 ± 30 mL/minute/1.73 m^2 for healthy young men and 126 ± 22 mL/minute/1.73 m^2 for healthy young women. Although there is no standard definition for increased GFR, an increase of 10% above the upper end of normal (greater than 160 mL/minute/1.73 m^2 in men and greater than 150 mL/minute/1.73 m^2 in women) has been proposed (Crit Care 2013;17:R35).
   b. ARC – Conditions such as surgery, trauma, burns, and early sepsis have been associated with increased renal blood flow. This is usually believed to be caused by an increased cardiac output coupled with vasodilation. The resulting ARC is believed to be a response to an inflammatory insult (e.g., systemic inflammatory response syndrome).
      i. One study found glomerular hyperfiltration in 17.9% of patients admitted to the ICU. Most of these patients were younger and admitted for multi-trauma or surgery (Anaesth Intensive Care 2008;36:674-80). These data are supported by a study showing young, postoperative trauma patients with peak creatinine clearance (CrCl) values as high as 190 mL/minute/1.73 m^2. A more recent study found significant risk factors for the ARC to be age 50 or younger, trauma, and a modified sequential organ failure score of 4 or less. Finally, a study of 133 trauma patients found age 56 or younger, SCr less than 0.7 mg/dL, and male sex to be independent risk factors for ARC. From this, study investigators developed a model to predict ARC in patients, the ARC in trauma intensive care (ARCTIC) score. A score of 6 or higher had a sensitivity of 0.843, a specificity of 0.682, a positive predictive value of 0.843, and a negative predictive value of 0.682.
      ii. A study of burn patients found an increase in iohexol clearance with a median value of 155 mL/minute/1.73 m^2 on day 1 of admission. In this small study, clearance had returned to the expected baseline of 122 mL/minute/1.73 m^2 by day 7 (Burns 2010;36:1271-6). In addition, several studies have shown increased excretion of renally eliminated drugs in burn patients, likely related to the second phase (hyperdynamic) of burn injury, which generally begins 48 hours after the burn injury. Examples include vancomycin, ciprofloxacin, imipenem, fluconazole, and aminoglycosides.
      iii. Fluid administration would be expected to improve cardiac output and thus renal blood flow. Animal studies have confirmed that the administration of crystalloids can transiently increase CrCl. Sheep administered normal saline and 3% hypertonic saline have a significantly higher calculated CrCl than controls. However, no human studies have verified ARC in critically ill patients after fluid administration.
      iv. Vasoactive drugs would be expected to improve cardiac output and thus renal blood flow. However, studies of humans did not corroborate this expectation and, instead, were only able to establish an improvement in CrCl in a subset of patients with impaired CrCl before norepinephrine administration. Although studies were unable to show development of ARC, patients receiving vasopressors for shock states might be expected to have normal renal blood flow and CrCl, assuming they are not experiencing AKI. The duration of ARC is not well established, but the peak CrCl appears to occur at about days 4–5 in most studies, with CrCl returning to normal by day 7 in one study.

c. Impaired renal clearance
   i. Decreased renal excretion of drugs during AKI is the most widely applicable change occurring in critically ill patients. Depending on the patient population and definition used, the incidence of AKI in ICU patients can be as high as 78%. AKI significantly affects the excretion of renally eliminated drugs, and dosing modifications must be made in these situations to avoid potential toxicity.
   ii. The Kidney Disease: Improving Global Outcomes (KDIGO) clinical practice guidelines for AKI recommend that staging of AKI be done using the KDIGO AKI criteria. However, these guidelines have no specific recommendations regarding drug dosing.
   iii. A clinical update to the 2010 KDIGO guidelines does recommend how to approach drug dosing in critically ill patients with AKI. Because of the complicated picture of AKI in critically ill patients, however, these recommendations are not as precise as the recommendations for drug dosing in CKD. In fact, the authors note that most renal dose adjustment recommendations in the literature and from the FDA (U.S. Food and Drug Administration) are based on data from patients with CKD (Kidney Int 2011;80:1122-37).
   iv. The update recommends a stepwise approach to adjusting drug-dosing regimens in patients with AKI (Box 2).

Box 2. Recommended Steps for Assessing and Adjusting Drug Regimens in Patients with AKI

Step 1 – Assess the following
Demographic information
Medical history (including history of renal disease)
Current clinical information
Current laboratory information
DNA polymorphisms
Step 2 – Estimate GFR (use the best equation according to patient factors)
Age
Body size
Ethnicity
Concomitant diseases
Step 3 – Review current medications
Identify drugs needing individualized dosing
Step 4 – Calculate individualized treatment regimen
Determine treatment goals (PK or PD values)
Calculate dosage regimen (according to drug PK and changes noted in the patient)
Step 5 – Monitor regimen
Drug response
Signs or symptoms of toxicity
Drug concentrations (if available)
Step 6 – Revise regimen
Adjust regimen according to patient response
Adjust regimen according to changes in patient status

AKI = acute kidney injury; GFR = glomerular filtration rate; PD = pharmacodynamic(s); PK = pharmacokinetic(s).

v. The recommendations include using the estimated GFR (eGFR) or CrCl to assess renal function for drug dosing. The update provides several equations that can be used to estimate kidney function, including the Cockcroft-Gault, Modification of Diet in Renal Disease (MDRD), and Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equations with a stronger emphasis on CKD-EPI over MDRD by the National Kidney Foundation. Recent evidence has suggested the removal of race adjustment in CKD-EPI, and MDRD has improved bias, accuracy, and precision of eGFR equations in Black adults (Int J Nephrol 2021;2021:1880499). The CKD-EPI equation can be calculated as follows:

$$GFR_{\infty} = 142 \times \min(SCr / \kappa, 1)^{\alpha} \times \max(SCr/ \kappa \ 1)^{-1.2} \times 0.9938^{Age} \times 1.012 \text{ (if female)}$$

where SCr is serum creatinine (mg/dL), κ is 0.7 for females and 0.9 for males, α is −0.241 for females and −0.302 for males, min indicates the minimum of SCr/κ or 1, and max indicates the maximum of SCr/κ or 1 (Am J Kidney Dis 2022;80:691-6).

vi. The Cockcroft-Gault equation is used to estimate the CrCl as follows:

$$CrCl = \frac{(140 - age)\,weight}{SCr \times 72} \times 0.85 \text{ (if female)}$$

where CrCl is in milliliters per minute, weight is in kilograms, and SCr is in milligrams per deciliter.

vii. The MDRD equation is described as follows:

$$GFR = 175.6 \times SCr/88.4^{-1.154} \times age^{-0.203} \times 0.742 \text{ (if female)}$$

where GFR is in milliliters per minute/1.73 m^2, SCr is measured in the laboratory using isotope dilution mass spectroscopy (IDMS), and age is in years.

viii. The update also notes that the most important factor when determining kidney function is having at least one GFR estimate for all patients.

3. Secretion and reabsorption can play important roles in drug concentration and clearance. However, it is difficult to study changes in drug secretion and reabsorption in patients. Therefore, data are not available to describe the clinically important changes in these two variables in critically ill patients.
4. Renal replacement therapies (RRT): See the Acute Kidney Injury chapter for more information regarding RRT and drug dosing.
   a. Patients receiving renal replacement therapy with a diagnosis of AKI may require hemodialysis and/or hemofiltration. The choice of dialytic technique depends on the institution, clinician expertise, patient hemodynamic stability, profile, and access to various types of renal replacement machines. Drug removal by dialysis depends on dialytic method.
   b. Acute intermittent hemodialysis: Intermittent hemodialysis can significantly contribute to the elimination of drugs, whereas other drugs are not appreciably removed by hemodialysis. Removal of drugs during hemodialysis depends on the size of the molecule, Vd, protein binding, and type of dialysis filter (specifically the membrane size).
   c. Continuous renal replacement therapies (CRRT): CRRT refers to several different methods of renal replacement therapy. Many studies have investigated the effect of CRRT on drug removal. Considerable variability exists in the type of CRRT used. The 2010 clinical update to the KDIGO guidelines suggests the following equation as one option to determine the appropriate drug dose in CRRT:

$$\text{dose} = \text{dose}_n \left(\frac{CL_{nonrenal} + (Q_{eff} \times SC)}{CL_{norm}} \right)$$

where dose is the desired dose for CRRT, dose n is the normal dose of a drug, $CL_{nonrenal}$ is the nonrenal clearance of a drug, Q_{eff} is the effluent rate, SC is the sieving coefficient, and CL_{norm} is the normal clearance of the drug. As the effluent rate (in milliliters per minute) increases, so does solute clearance. Therefore, it is the most important CRRT parameter pertaining to drug removal.

B. Hepatic Excretion – Hepatic excretion of drugs is less important for most drugs than renal excretion. However, excretion of drug in the bile can potentially be affected by critical illness. This is evidenced by changes in the clearance of some neuromuscular blocking agents.
   1. A study of nine patients undergoing surgery for total biliary obstruction showed a significant increase in pancuronium half-life compared with normal patients (270 minutes vs. 132 minutes, $p<0.001$). The urinary excretion of pancuronium and its metabolites did not change. This suggests that the increased half-life was caused by the decreased hepatic excretion of pancuronium (Br J Anaesth 1977;49:1103-8).
   2. Results were similar for vecuronium in patients with cholestasis, in which the mean half-life was 98 minutes in patients with cholestasis and 58 minutes in normal patients (Br J Anaesth 1986;58:983-7).

C. Pulmonary Excretion – Pulmonary excretion is important for volatile gases such as anesthetics. It can be hypothesized that impaired gas exchange (e.g., acute respiratory distress syndrome) affects the body's ability to remove volatile gases. However, data are lacking regarding changes in critically ill patients that affect their ability to excrete anesthetics.

VII. PHARMACODYNAMICS

A. *Pharmacodynamics* refers to the biochemical and physiologic effects of a drug, specifically those related to the mechanism of action.

B. This term also pertains to drug/receptor binding and clinical effect. Most clinicians use the term to refer to the physically identifiable effect of a drug in a patient. For example, the PD effect of an opioid is the relief of pain reported by the patient. However, the PD effect of some drugs is not readily observable. For example, the PD effect of proton pump inhibitors is an increase in gastric pH. Few PD studies of critically ill patients are reported in the medical literature, and most pertain to antibiotic therapy.
   1. Antibiotic overview: PD studies of antibiotics use models to estimate the combined effects of the patient population PK of specific antibiotics and the MIC for select bacteria. These techniques usually allow a calculation of the desired PD outcome. Antibiotics generally fall into three PD categories, which correlate with efficacy: (1) time-dependent killing (T>MIC), (2) concentration-dependent killing (C_{max}/MIC), and (3) a combination of time- and concentration-dependent killing (ratio of area under the curve to the minimum inhibitory concentration for the bacterial pathogen [AUC/MIC]).
   2. β-Lactam antibiotics: For β-lactam antibiotics, the PD parameter of the free drug concentration time above the MIC (*f*T>MIC) is used to predict treatment success. This is reported as a percentage of time the free drug concentration remains above the MIC. The ideal *f* T>MIC is 100%. However, PD studies of β-lactam antibiotic use in critically ill patients have found that a low percentage of patients will achieve the desired PD targets (Crit Care 2011;15:R206). These failures are often attributed to clinically important changes that can occur rapidly in critically ill patients (e.g., ARC). As such, epidemiologic studies have tried to determine a breakpoint at which clinical success is achieved. Studies vary, and the suggested breakpoint is 50%–100% (*f*T>MIC) (Br J Clin Pharmacol 2012;73:27-36), with others

suggesting cutoffs of about 40% for carbapenems, 50% for penicillins, and 50%–75% for cephalosporins (Clin Infect Dis 2008;26:1-10). Modeling usually suggests improved PD of β-lactams when using prolonged or continuous infusions. Many institutions have adopted the practice of prolonged infusions. This is supported by quasi-experimental and retrospective studies showing improved outcomes such as improved clinical cure, improved microbiological cure, and reduced morbidity and mortality. However, prospective controlled clinical trials have had mixed results (J Crit Care 2014;29:1089-95; Am J Respir Crit Care Med 2015;192:1298-305). There are many reasons why there appears to be a discrepancy between PD modeling studies and controlled clinical trials; patient variability, dosing variability, and disease severity seem to be important factors (Ann Intensive Care 2012;2:37). Another variable to consider is resolving AKI and timely dose adjustment of β-lactam antibiotics because delayed dose adjustments have been associated with in-hospital mortality (Open Forum Infect Dis 2024;11:ofae059).

3. Aminoglycosides: Aminoglycoside bacterial killing is based on the ratio between the maximum drug concentration and the MIC for the bacterial pathogen (C_{max}/MIC), or concentration-dependent killing. Efficacy was noted when patients were pooled from four controlled clinical trials and peak concentrations, MIC values, and clinical response were evaluated. Peak-to-MIC ratios of 8–10 resulted in around a 90% clinical response (J Infect Dis 1987;155:93-9). In patients with gram-negative bacteremia, early therapeutic peak concentrations were a significant discriminating factor for mortality. According to these and other data, once-daily aminoglycoside dosing has been used. Taking advantage of high peak concentrations maximizes the PD of aminoglycosides. Variability in the Vd of aminoglycosides in critically ill patients, together with concern for ARC in this population, raises issues about appropriately dosing these agents, especially in critically injured trauma patients, whose drug concentrations can be undetectable for more than 12 hours (J Trauma 2000;49:869-87) and larger loading doses and more frequent monitoring may be warranted.
4. Vancomycin: The PD parameter that best describes vancomycin efficacy is the AUC/MIC. Several studies have evaluated the free 24-hour AUC/MIC ($fAUC_{0-24}$/MIC), or the AUC × 50% protein binding/MIC. Current guidelines use the available literature to recommend an AUC/MIC of 400 or greater for effectiveness and less than 600 for safety. (Am J Health-Syst Pharm 2020;77:835-63). Loading doses of vancomycin of 20–35 mg/kg based on actual body weight are recommended to expedite achieving these target serum concentrations and PD goals while decreasing the chance of subtherapeutic serum concentrations during the initial days of therapy. The guidelines recommending continuous infusion regimens are an acceptable alternative when conventional dosing is unable to achieve the target AUC. They also note the intravenous access issues presented for critically ill patients when continuous infusion regimens are employed (Clin Infect Dis 2020:71;1361-4). Specifically, vancomycin has been demonstrated to be Y-site incompatible with multiple β-lactams, moxifloxacin, propofol, phenytoin, methylprednisolone, and furosemide—all medications that are commonly used in the ICU setting. As such, clinicians should closely monitor vancomycin coadministration with other medications. Of interest is a retrospective study of vancomycin-associated nephrotoxicity in critically ill patients. In this study, intermittent dosing was associated with a significantly higher risk of nephrotoxicity than continuous infusion (odds ratio 8.2; $p<0.001$) (Crit Care Med 2014;42:2527-36). Of note, more aggressive dosing may be required in critically ill patients. Doses as high as 20 mg/kg administered as often as every 6 hours were needed to optimize PK variables in critically injured trauma patients being treated for ventilator-associated pneumonia (J Trauma Acute Care Surg 2012;72:1478-83).
5. Fluoroquinolones: Similar to the efficacy of aminoglycosides, the efficacy of the fluoroquinolones is based on a C_{max}/MIC (10 or greater), and they exhibit a post-antibiotic effect against gram-negative and gram-positive bacteria. PD studies have shown that the $fAUC_{0-24}$/MIC is associated with bacterial eradication. In one study of lower respiratory tract infections treated with ciprofloxacin, an AUC_{0-24}/MIC of 125 was associated with the percent probability for clinical cure of 80% (Antimicrob Agents Chemother 1993;37:1073-81). However, it is difficult to incorporate these PD variables into fluoroquinolone dosing in individual critically ill patients. Measuring fluoroquinolone serum concentrations is not routine; therefore, it is difficult to determine whether the PD targets have been met in an individual patient.

6. Nonantibiotic drugs: PD studies of other drugs in critically ill patients are sparse.
    a. The PD parameter for continuous infusions of many anticoagulants is the change in activated partial thromboplastin time. Unfractionated heparin infusions are predictable in most patient populations. However, in critically ill patients, less than one-half of patients (44%) reached therapeutic activated partial thromboplastin times within 24 hours of heparin infusion initiation (Neth J Med 2013;71: 466-71). Concern for a variable response in critically ill patients has led to the development of modified dosing nomograms/protocols. Researchers have found a shortened time to therapeutic activated partial thromboplastin times in critically ill patients receiving unfractionated heparin and direct thrombin inhibitors (argatroban and bivalirudin) with nomograms.
    b. As with antibiotic PD studies, most PD studies of other drugs have shown a decreased response in critically ill patients. For example, critically ill patients in septic shock had a reduced response to dobutamine compared with those without septic shock and with normal volunteers (Crit Care Med 1993;21:31-9). Trauma patients with edema have lower AUCs for anti-Xa activity than do non-edematous patients (J Trauma 2005;59:1336-43). Mechanically ventilated patients with chronic obstructive pulmonary disease were studied for covariates affecting acetazolamide therapy. Mixed-effects modeling found the Simplified Acute Physiology Score II, serum chloride concentrations, and concomitant corticosteroids to be the main covariates interacting with acetazolamide PD.

VIII. THERAPEUTIC DRUG MONITORING

A. Therapeutic drug monitoring (TDM) refers to the measurement of drug concentrations in the blood. The focus of TDM is on drugs with a narrow therapeutic index and aims to achieve two goals: (1) maximize efficacy and (2) reduce toxicity. Use of TDM in critically ill patients is extremely important because changes in the PK variables previously described can result in less-than-desirable drug concentrations. Table 2 highlights commonly used medications in the ICU and their corresponding therapeutic ranges. One of the main limitations of TDM is the lack of clinically available assays. In addition, assays for some drugs may not be sufficiently cost-effective to routinely conduct in certain institutions. In addition, assays for some drugs may not be sufficiently cost-effective or timely, thus limiting their routine use in some institutions. These issues usually result in TDM being available for a very limited spectrum of drugs.
    1. Monitoring of blood concentrations depends on the intended use and interpretation of those concentrations. Most TDM occurs as a method to confirm a therapeutic concentration in a patient with signs and/or symptoms of toxicity or decreased efficacy. In this case, a concentration is measured during the appropriate time interval (Table 2), and a clinician interprets the concentration. If needed, the clinician modifies the drug dosing according to clinical experience. This method may produce variable results. Critically ill patients require important considerations. For example, if extended-interval dosing is being used, the likelihood of an increased Vd must be considered. Patients with ARC have the potential to have a prolonged drug-free period. Finally, the status of a critically ill patient can change rapidly. Monitoring for decreased kidney function is essential to avoid accumulation.

Table 2. TDM Ranges for Select Drugs Used in Critically Ill Patients

Drug	Timing of Blood Sample	Therapeutic Range[a]	Unbound Therapeutic Range
Amikacin	Peak (extended interval) Trough (extended interval) Peak (traditional) Trough (traditional)	30–40 mg/L undetectable 20–30 mg/L < 8 mg/L	—
Carbamazepine	Trough	4–12 mg/L	0.5–4 mcg/mL
Cyclosporine	Trough	50–500 mcg/L	—
Digoxin	Trough (8–24 hr postdose)	0.6–2 mcg/L	0.4–0.9 ng/mL
Gentamicin	Peak (extended interval) Trough (extended interval) Peak (traditional) Trough (traditional)	5–20 mg/L undetectable 8–10 mg/L < 2 mg/L	—
Lidocaine	Peak	1.5–5 mg/L	—
Phenobarbital	Trough	10–40 mg/L	—
Phenytoin	Trough	10–20 mg/L	1–2.5 mcg/mL
Tobramycin	Peak (extended interval) Trough (extended interval) Peak (traditional) Trough (traditional)	5–20 mg/L undetectable 8–10 mg/L < 2 mg/L	—
Valproate	Trough	50–100 mcg/mL	5v15 mcg/mL[b]
Vancomycin	Peak and Trough to calculate AUC	AUC 400–600 mg × hour/L	—

[a]Therapeutic ranges for antibiotics may be modified on the basis of pharmacodynamic target attainments (e.g., peak to MIC ratio).

[b]Optimal range for unbound valproate remains undetermined.

TDM = therapeutic drug monitoring.

Data adapted from: J Antimicrob Chemother 2016;71:2754-9; Clin Infect Dis 2020:71;1361-4; Ther Drug Monit 2017;39:522-30; Infect Dis 2023;10:ofad058; Pharmacotherapy 2008;28:1391-400; Am J Cardiol 2012;109:1818-21.

2. For certain intravenously administered drugs (e.g., aminoglycosides, vancomycin), several concentrations can be measured when the drug is at steady state. The patient's specific PK can be determined and used to tailor the drug-dosing regimen. The following PK equations can be used to calculate the various kinetic variables.

   Determination of the elimination rate constant:

$$k_e = \frac{\ln \frac{C_1}{C_2}}{\Delta \text{ in time}}$$

   where k_e is the elimination rate constant, C_1 is the measured peak concentration, C_2 is the measured trough concentration, and Δ in time is the elapsed time from C_1 to C_2 in hours.

   Determination of the drug half-life:

$$t_{1/2} = \frac{0.693}{k_e}$$

   where $t_{1/2}$ is the calculated half-life.

Determination of the calculated peak concentration:

$$C_{max} = C_1 (e^{ke(t)})$$

where C_{max} is the calculated peak concentration and t is the time between the C_1 and the end of the intravenous infusion in hours.

Determination of the calculated trough concentration:

$$C_{min} = C_2 (e^{-ke(t)})$$

where C_{min} is the calculated trough concentration and t is the time between C_2 and the beginning of the next dose.

Determination of the drug Vd:

$$Vd = \frac{dose}{t_{inf} \times k_e} \times \frac{(1 - e^{-ke(t_{inf})})}{C_{max} - C_{min} \times e^{-ke(t_{inf})}}$$

where t_{inf} is the duration of the drug infusion.

Determination of the new dosing interval:

$$\tau = \frac{\ln\left(\frac{C_{max,desired}}{C_{min,desired}}\right)}{k_e} + t_{inf}$$

where τ (tau) is the new dosing interval, $C_{max,desired}$ is the desired peak concentration for the new dosing regimen, and $C_{min,desired}$ is the desired trough concentration for the new dosing regimen.

Determination of the new dose:

$$dose = C_{max,desired} \times k_e \times Vd \times t_{inf} \times \frac{(1 - e^{-ke(\tau)})}{(1 - e^{-ke(t_{inf})})}$$

3. One drawback of this method is that it assumes that the PK variables obtained (e.g., vancomycin trough) correlate with PD variables (AUC/MIC). Although this may be true in many cases, some advocate incorporating PD into individualized drug dosing. Alternative methods use PK variables from previous patients (population PK) to estimate the PK in an individual patient. These methods usually require complicated mathematical calculations, many of which are not practical for clinical use. Many software programs are now available for clinician use in patient care that simplify the process of carrying out complex calculations. Bayesian modeling has also been proposed to address the issues posed when using population PK in patient groups that may not be well represented in the population. Using software with population PK variables from non-critically ill patients to interpret or predict PK in critically ill patients could result in errors in designing an appropriate drug-dosing regimen.
4. A position paper provides a more detailed review of the PK, PD, and TDM of commonly used antimicrobials in critically ill patients and notes recommendations that certain antimicrobials should be monitored with TDM (Intensive Care Med 2020;46:1127-53). The aminoglycosides and vancomycin are included on this list because they have routinely been monitored in critically ill patients. The recommendations also include the β-lactam class of antibiotics, linezolid, and the antifungal voriconazole. Finally, the document provides PK and PD target variables for these drugs. The greatest barrier continues to be access to local laboratory analysis and results within a clinically relevant time frame. The reader is referred to the position paper for a more in-depth review.

B. Drug Nomograms/Protocols – A nomogram is a diagram representing the relationship between three or more variables using scales arranged such that one variable can be determined if the other variables are known. A classic drug-dosing example is the once-daily aminoglycoside nomogram (Antimicrob Agents Chemother 1995;39:650-5). The nomogram allows the user to determine only one variable (usually, the dosing interval) using other variables (time from infusion to drug concentration and the measured drug concentration). Many institutions incorporate nomograms into drug-dosing protocols or pathways and routinely refer to them as *drug-dosing nomograms*. Of note, many of these nomograms (e.g., Hartford) were not developed in critically ill patients and likely do not consider the potential for alterations in PK. One of the most commonly used drug-dosing nomograms was developed for continuous infusions of unfractionated heparin. As previously noted, a heparin nomogram can improve time to therapeutic activated partial thromboplastin time when developed specifically for critically ill patients. However, the patient population used to develop a nomogram should be considered because the variables may not apply to all patient populations. An example of this occurs with protocols for insulin infusions in critically ill patients. As previously noted, patients with AKI are at a greater risk of hypoglycemia if treated with an insulin infusion protocol developed in critically ill patients without kidney impairment.

IX. CONCLUSION

There are marked differences in the ways in which critically ill patients respond to drugs. Research in this area has noted significant changes in the PK and PD of certain medications in select critically ill populations. Although these studies have highlighted important issues, considerable work is still needed to better define these changes in different critically ill populations. As research continues to advance, together with our knowledge of how patients respond to drugs differently, critical care clinicians must stay abreast of new information and the ways in which it will affect the care of their patients.

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ANSWERS AND EXPLANATIONS TO PATIENT CASES

1. **Answer: B**
Increasing the concentration of an antibiotic at an infection site is most important (Answer B is correct). Although case reports describe the use of intraventricular antibiotics in the treatment of meningitis, they have not shown superiority (Answer A is incorrect). Although there is the potential to reduce vancomycin-induced nephrotoxicity, no studies have compared the nephrotoxicity of intraventricular antibiotics with that of intravenous antibiotics, likely because this is not the rationale for their use (Answer C is incorrect). Ototoxicity could also be reduced, but this was not the intent of the locally instilled antibiotics (Answer D is incorrect).

2. **Answer: D**
Three studies have documented an increase in digoxin absorption when the gastric pH is increased, with two studies noting the cause of increased gastric pH from a proton pump inhibitor (Answer D is correct). However, no studies have shown a decreased absorption of carvedilol (Answer A is incorrect), ciprofloxacin (Answer B is incorrect), or diazepam (Answer C is incorrect).

3. **Answer: A**
A decrease in the ionization of a drug allows the drug to pass more easily through membranes. Increasing the ionization would decrease the Vd by decreasing its ability to pass through membranes. A weak acid would be less ionized in a more acidic environment. The patient is likely in diabetic ketoacidosis with a definite acidosis. Aspirin (a weak acid) is less likely to be ionized and would have an increased Vd (Answer A is correct; Answers B–D are incorrect).

4. **Answer: C**
Levofloxacin has a large Vd. The increase in interstitial fluid volume caused by 6 L of crystalloids would increase the Vd of hydrophilic drugs but would not appreciably affect the Vd of a drug like levofloxacin, which already has a large Vd (Answer C is correct). Tobramycin, daptomycin, and cefepime have relatively small Vd values and are increased in patients with increased interstitial volumes (Answers A, B, and D are incorrect).

5. **Answer: B**
Morphine is a high extraction ratio drug. Mechanical ventilation can decrease cardiac output and thus decrease liver blood flow. The decrease in liver blood flow is inversely proportional to the unbound steady-state concentration. Therefore, it both decreases the hepatic blood flow and increases the unbound concentration (Answer B is correct). The effects of mechanical ventilation on cardiac output would likely decrease oxygen delivery (Answer A is incorrect). Cytokines affect intrinsic clearance but would not affect the concentration of a high extraction ratio drug (Answer C is incorrect). Mechanical ventilation can indirectly affect the unbound concentration (Answer D is incorrect).

6. **Answer: B**
This patient has sepsis, which is associated with an increased production of inflammatory cytokines. These cytokines can decrease the activity of the CYP enzymes and decrease intrinsic clearance. In addition, fluconazole inhibits CYP2C19 activity. Diazepam is a low extraction ratio drug. The hepatic clearance of diazepam is affected by changes in intrinsic clearance (including CYP2C19 activity). The unbound steady-state concentration would be increased, because of a decrease in intrinsic clearance caused by the inflammatory cytokines and CYP2C19 inhibition by fluconazole. The increased unbound steady-state concentrations would cause increased sedation (Answer B is correct). There is no cause for an increase in intrinsic clearance and a corresponding decrease in sedation (Answer A is incorrect). There is no reason for this patient to have a decrease in the unbound fraction of diazepam (Answer C is incorrect). A decrease in albumin could occur with sepsis and would result in an increase in the unbound fraction of diazepam. The increase in unbound fraction would not affect the unbound steady-state concentration, nor would it affect the level of sedation (Answer D is incorrect).

7. **Answer: D**
Under normal conditions, the unbound steady-state concentration for phenytoin is expected to be 10% of the total concentration. Thus, a total concentration of 10-20 mg/L is equivalent to an unbound steady-state concentration of 1-2 mg/L. When the unbound fraction changes, this ratio changes. The unbound steady-state

concentration does not change, but the total concentration decreases, which causes the change in the ratio. Using the equations for low extraction ratio drugs (like phenytoin) to explain what has occurred, the unbound fraction is inversely proportional to the total concentration but has no effect on the unbound steady-state concentration (Answer D is correct; Answers A, B, and C are incorrect).

8. **Answer: A**

The kidney plays a role in the excretion and metabolism of insulin. Acute kidney injury will decrease the ability of the kidney to metabolize insulin, which will increase circulating insulin and contribute to hypoglycemia (Answer A is correct). The Vd of insulin in AKI is not well studied and has not been linked with episodes of hypoglycemia (Answer B is incorrect). Hepatic impairment may increase the risk of hypoglycemia with insulin, but an increase in hepatic metabolism is unlikely to affect the risk of hypoglycemia in AKI (Answer C is incorrect). Insulin resistance has been reported in patients with AKI, but this would not contribute to hypoglycemia (Answer D is incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: B**

Decreased anti-Xa activity occurs in critically ill patients receiving several different low-molecular-weight heparins. This patient received the standard fluid bolus and then required more fluid to raise his central venous pressure. This suggests that the patient has redistributed the fluids to the extravascular space. The data showing that edematous patients have lower anti-Xa activity than non-edematous patients supports the connection between anti-Xa activity and absorption (Answer B is correct). Given the patient's fluid distribution, the Vd of enoxaparin would likely be increased, not decreased (Answer A is incorrect). Enoxaparin is not hepatically metabolized (Answer C is incorrect). Although a patient could have an increased anti-Xa activity secondary to decreased renal elimination, this would require a CrCl of less than 30 mL/minute/1.73 m^2. That the patient had a deep venous thrombosis does not correlate with increased activity (Answer D is incorrect).

2. **Answer: C**

Abdominal surgery has been identified as a risk factor for ileus. Antisecretory agents such as pantoprazole alter drug absorption. Traumatic brain injury is significantly associated with intolerance of enteral nutrition, as indicated by increased gastric residuals. This indicates delayed gastric emptying and the risk of altered absorption. Therefore, the combination of abdominal surgery, pantoprazole therapy, and TBI contains the three variables identified in the literature to alter absorption (Answer C is correct). Theoretically, intestinal atrophy could cause changes in absorption, but no data are available to confirm this theory (Answers A and D are incorrect). Changes in cardiac output have been correlated with changes in hepatosplanchnic blood flow. These changes in blood flow are thought to affect absorption, but again, no data have correlated increased cardiac output with increased drug absorption (Answer B is incorrect).

3. **Answer: B**

Propofol is a high extraction ratio drug that is bound to albumin. When the albumin concentration decreases, there is an expected increase in the free fraction of propofol. Using the equations describing the total and unbound concentrations of a high extraction ratio drug, the total concentration is not affected by changes in the free fraction, but unbound concentrations are increased when the free fraction increases (Answer B is correct; Answers A, C, and D are incorrect).

4. **Answer: C**

The update to the KDIGO guidelines notes that the most important factor in determining kidney function is having at least one estimate of GFR. The update recommends that the GFR or the CrCl be estimated to determine this (Answer C is correct). The BUN value can help identify the BUN/SCr ratio indicating intravascular volume contraction and, potentially, a prerenal cause to the patient's AKI, but it does not help in drug dosing (Answer A is incorrect). Total daily urinary output is helpful in diagnosing AKI (oliguric vs. non-oliguric AKI) and in staging AKI (using a urinary output of less than 0.5 mL/kg/hour) but not in drug dosing (Answer B is incorrect). A history of CKD can help determine a baseline kidney function, but the GFR is still needed for changes in drug dosing (Answer D is incorrect).

5. **Answer: C**

The ASHP (American Society of Health-System Pharmacists) vancomycin dosing guidelines recommend a 20- to 25-mg/kg loading dose of vancomycin for serious infections. Given the patient's sepsis, he is likely to have more interstitial fluid, which will increase the Vd of hydrophilic drugs like vancomycin. The 2500-mg dose is near the top end of the recommended 20- to 25-mg/kg loading dose (Answer C is correct). The other doses are outside the recommended 20- to 25-mg/kg loading dose range (Answers A, B, and D are incorrect).

6. **Answer: A**

Critically ill patients younger than 50 are more likely to have augmented renal excretion. Vancomycin is excreted unchanged in the urine. The initial dosing would result in a therapeutic trough concentration, given the increased (augmented) renal excretion of vancomycin. Augmented renal excretion usually returns to normal around day 7. An increased vancomycin trough after a previously therapeutic trough is most likely associated with decreased renal excretion at day 10 of therapy (Answer A is correct). There is no indication

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

that the Vd has changed, which would likely result in lower or unchanged concentrations (Answer B is incorrect). Vancomycin tissue penetration may be better if inflammation is present, but no documented studies correlate decreased inflammation with increased serum concentrations (Answer C is incorrect). The liver does not appreciably metabolize vancomycin, and liver blood flow would not be a factor in the vancomycin concentration (Answer D is incorrect).

7. **Answer: C**

Prospective controlled studies of prolonged piperacillin/tazobactam infusions have shown no improvement in mortality (Answer A is incorrect). No studies have reported neurotoxicity as an outcome (Answers B and D are incorrect). Only retrospective studies have shown improvements in mortality with the use of prolonged or continuous infusions of piperacillin/tazobactam (Answer C is correct).

8. **Answer: C**

Morphine is a high extraction ratio drug. As such, its hepatic metabolism or hepatic clearance depends only on hepatic blood flow. Hepatic clearance equals hepatic blood flow (Answer C is correct). Morphine does not bind to AAG (Answer A is incorrect). Because hepatic blood flow is the main determinant of hepatic metabolism of morphine, changes in protein binding do not affect metabolism (Answer B is incorrect). Changes in intrinsic clearance do not affect the metabolism of morphine as significantly as does hepatic blood flow (Answer D is incorrect).